r/NoStupidQuestions

Why can't we use the heat generated by AI/data centers to generate electricity?

Posted by u/opus666 • • 0 points • 41 comments

Most means of electricity generation seems to be "steam makes turbine go brrrr"

Data centers have a lot of cooling requirements but why can't we harness that thermal energy to generate electricity? Does it just not get hot enough to boil water?

Comments

u/[deleted] • 1 points • 2026-03-19 03:22 UTC

[removed]

u/Old_Schedule_3667 • 1 points • 2026-03-19 03:13 UTC

Knowing Lifewood I see that using waste heat from data centers for district heating or industrial processes is an interesting idea, but it needs careful thermal management and infrastructure planning. For enterprise AI workloads, secure, scalable, and compliant data handling remains the main priority, and I wonder if there are opportunities for AI generated heat reuse in your context.

u/BarberProof4994 • 1 points • 2026-03-18 21:35 UTC

This is like asking why don't we use the heat in Florida to make electricity. After all it's hotter there

The heat needs to be hot enough to make steam and that steam needs to be under enough pressure to spin a turbine fast enough and long enough to generate electricity.

And the amount of electricity generated this way loses a LOT of usable energy.

For data centers, the way computers and chips are designed, you can't use that heat THERE, because the heat damages the components. So you'd have to move that what elsewhere.

And in that process you'd lose heat through radiation. So you'd have no possible way to get enough heat safely away from the at risk component and accumulated in one location at a hot enough temp to do anything.

u/MrZwink • 1 points • 2026-03-18 10:04 UTC

A better idea, is just to transport that heat and use it for something else. Build a datacentre next to a pool, use the waste heat to heat the pool.

u/LavishnessCapital380 • 1 points • 2026-03-18 07:15 UTC

Like 99.9% of electricity consumed is loss as heat.

u/farmthis • 1 points • 2026-03-18 05:28 UTC

Entropy... Really hard to capture and extract energy from low temperature differentials. Stirling engines--using the expansion and contraction of gas--is one of the most efficient ways but it doesn't scale particularly well.

u/Temporary-Ask-1129 • 1 points • 2026-03-18 05:24 UTC

Heat into electricity into heat into electricity. What you're talking about is so inefficient its not worth doing even if it could get up enough energy to boil water to steam again.

Better idea is to shut all that shit down before it gets worse.

u/Kokophelli • 1 points • 2026-03-18 05:00 UTC

Grow tomatoes

u/Wendals87 • 1 points • 2026-03-18 04:25 UTC

You need heat hundreds of degrees celcius

Datacentres don't get anywhere near that hot

u/breakerofh0rses • 1 points • 2026-03-18 02:37 UTC

You'd want something like a stirling engine and not a steam engine, and even then you're not capturing enough energy to make it worthwhile. I suppose it'd be possible to entirely redesign the cooling to where you could make it significantly better, but I strongly suspect that it would still be very net negative on energy. That said, if you could make it perform well enough it may be worth taking that hit to reduce both the power and water consumption of them.

u/JustinTimeCuber • 1 points • 2026-03-18 01:37 UTC

Technically possible, would be considered a form of regenerative cooling, but the efficiency would be too low to recover a meaningful amount of the input energy.

u/Responsible-Mall-991 • 2 points • 2026-03-18 01:30 UTC

Since it's low grade heat, as some already mentioned, a better application would be to use heat pumps for space or water heating, perhaps for a District Energy System

u/OldGeekWeirdo • 4 points • 2026-03-17 23:57 UTC

>Does it just not get hot enough to boil water?

You're on the right track. Think of heat-to-energy like a building a hydroelectric dam across a stream. It only works if the height difference between the front and back of the dam is enough to make enough power.

Using the heat from a data center would be like trying to build a dam with only 1' of height difference. You could create power, but not enough to really be useful.

u/Far_Gift6173 • 1 points • 2026-03-18 08:50 UTC

couldn't you heat some kind of garden with it to grow vegetables all year?

u/OldGeekWeirdo • 1 points • 2026-03-18 17:57 UTC

I'd think that's a possibility. But you'd have the expense overhead of a building. Is that going to be cost competitive with produce brought in from the fields from other parts of the country/world? I suspect not. It would have to compete on the local/farm fresh angle.

u/Far_Gift6173 • 1 points • 2026-03-18 18:07 UTC

I don't know. Bu I think that there is seasonal vegetables that you could grow all year with it.

The netherlands seem to be trying to do this, but this is probably just a testballoon to see whether it's worth it

https://www.ri.se/en/greenhouses-can-be-heated-by-waste-heat-from-data-centres

u/Alita-Gunnm • 1 points • 2026-03-18 01:15 UTC

It would also make it harder and more expensive to cool the data center.

u/OldGeekWeirdo • 2 points • 2026-03-18 17:50 UTC

True. For dams to work, you have to block the water flow and let it back up. That's going to put a strain on the data center cooling system.

But using the waste heat for lower temperature uses like heating houses would work.

u/groundhogcow • 3 points • 2026-03-17 22:33 UTC

There is a guy who is using the heat to run a 3d printer.

the heat level is to low to generate electricity.

Another guy was using the heat to heat a public pool.